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Tailored Well Completion and Hydraulic Fracturing Design Workflow to Account for Heterogeneity in Well Geomechanical and Petrophysical Properties

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Abstract

In developed tight reservoir fields, operators typically anticipate that new wells will perform similar to offsets. When performance deviates sharply, whether underperforming or overperforming, it can be puzzling, especially when field development planning has long relied on consistent offset data. These deviations often point to localized geomechanical and petrophysical heterogeneities that are not recognized early. Without targeted diagnostics before completion and hydraulic fracturing design, the resulting stimulation program may fail to create optimal fracture geometry, compromising production potential or misuse of resources.

This paper presents three field cases from distinct geological and operational contexts an Indian offshore tight gas field, Argentina's Vaca Muerta Shale, and Saudi Arabia's deep formations where late-stage identification of localized anomalies or completion constraints directly influenced well outcomes and design decisions. [Table 1](#) summarizes the key localized conditions, design implications, and results from each case: Case 1 shows how high fracture initiation pressure and low brittleness demanded substantially larger stimulation volumes and higher rates (Dutta et al., 2019); Case 2 demonstrates how low-stress conditions enabled full SRV development with smaller, lower-rate treatments (Altman et al., 2018); and Case 3 illustrates how HPHT and sour service environments shifted the limiting factor from reservoir quality to completion hardware integrity (Solares, 2001).

Table 1—Summarized Key Insights from the Three Case Studies

Case Study	Reservoir & Location	Key Challenge	Diagnostics Used	Main Findings	Solutions Implemented	Results / Impact
1 – Reliance Industries, India	Tight gas, multiple depth intervals	Deepest wells underperformed despite highest TOC & HCFP	Core analysis, sonic-density cross-plots, mineralogy trends	Clay diagenesis & fluid expansion increased FIP, reduced brittleness	Revised frac design; stress profile recalibration	Improved fracture geometry, partial recovery in productivity
2 – Vaca Muerta Shale, Argentina	Unconventional shale	Best-performing well had lowest HFP	Log composite analysis, Shmin mapping, fracture simulation	Low Shmin enabled longer fractures & larger SRV despite low HFP	Targeted completion design optimizing stage length	Significantly higher EUR in low-HFP, low- Shmin well. lower cost.
3 - Saudi Arabia	Deep sour gas (HPHT)	Ultra-high FIP (~16,000 psi) & H ₂ S; completion mechanical limits	Tubing stress analysis, load case modeling	Connector & seal limitations under frac loads; piston effect critical	Upgraded materials (C- 95/ P-110), redesigned components	15 wells frac'd successfully with no failures

A multidisciplinary workflow is proposed and discussed in this paper to identify such well-specific anomalies before committing to completion and stimulation designs. The workflow begins with petrophysical and mineralogical profiling to quantify brittleness, hydrocarbon-filled porosity, and other frackability indicators. Geomechanical stress mapping and fracture initiation pressure prediction follow, enabling engineers to define a tailored stimulation program. Only after the frac design is established is completion hardware selected, ensuring the system can handle predicted loads without overstress. The sequence concludes with real-time monitoring and post-job feedback to refine field development models.

The lessons from these three contrasting cases demonstrate that even in mature fields, targeted early detection of localized heterogeneity or mechanical constraints can guide either the intensification of stimulation parameters in low-frackability zones, optimization through job downsizing in favorable stress environments, or hardware redesign in completion-limited situations. All three approaches improve capital efficiency, reduce operational risk, and enhance sustainability in technically challenging reservoirs.

Introduction

Development of tight and unconventional reservoirs has advanced rapidly over the last two decades, driven by improvements in hydraulic fracturing, horizontal drilling, and geomechanical characterization. However, despite similar completion designs and apparently favorable reservoir quality indicators such as total organic content (TOC) and hydrocarbon-filled porosity (HCFP), production results can vary sharply between wells in the same field. In many cases, these discrepancies stem from localized geomechanical or petrophysical variations that are not incorporated into the initial completion and stimulation design.

A common practice in mature and emerging basins is to extrapolate field development plans from a limited set of "offset" well results without systematically checking for subsurface variability. This approach can lead to high-potential wells underperforming due to overlooked mineralogical transitions, stress anisotropy, fracture initiation pressure (FIP) anomalies, or even completion hardware limitations. Such constraints from the latter can be especially critical in high-pressure, high-temperature (HPHT) and sour gas environments, where the mechanical limits of tubulars and connectors can restrict achievable fracture designs regardless of reservoir quality (Solares, 2001; Vagenes, 2012).

The three field cases presented in this paper highlight the consequences of late-stage identification of such localized anomalies. Case 1 (India) illustrates how high FIP and low brittleness required substantially higher stimulation volumes and rates (Dutta et al., 2019). Case 2 (Argentina) demonstrates how low minimum horizontal stress (Shmin) enabled large stimulated reservoir volume (SRV) with reduced pumping effort

despite lower HCFP (Altman et al., 2018). Case 3 (Saudi Arabia) shows how HPHT and H₂S service conditions shifted the limiting factor from reservoir quality to completion hardware integrity (Solares, 2001).

These examples underline the need for an integrated, multidisciplinary workflow linking petrophysical evaluation, geomechanical modeling, stimulation design, and completion mechanical verification to tailor well designs to specific geologic and operational realities. This paper proposes such a fit-for-purpose workflow, designed to detect localized heterogeneity early in the planning cycle, reduce costly redesigns, and improve both operational safety and capital efficiency.

Literature Review

The Importance of Integrating Petrophysical and Geomechanical Data

The variability of completion performance in unconventional and tight gas reservoirs is often linked to the degree of integration between petrophysical data and geomechanical characterization during planning. Previous studies (Barree et al. 2011; Molinas et al. 2019) have shown that rock mechanical properties, such as Young's modulus, Poisson's ratio, and unconfined compressive strength (UCS), can vary significantly within a single reservoir interval due to changes in lithology, diagenesis, and burial history. When fracture designs do not account for these variations, fracture propagation can be suboptimal, leading to reduced stimulated reservoir volume (SRV) and lower productivity.

Fracture Initiation Pressure (FIP) and Brittleness

Fracture initiation pressure (FIP) is a critical design parameter in hydraulic fracturing, influencing pump schedule, proppant selection, and equipment requirements.)Dutta et al., 2019 (notes that in some deep, overpressured gas wells, FIP can be significantly elevated due to bound water expansion in clays, cementation, or stress redistribution from tectonic activity. Brittleness indices, calculated from mineralogy and elastic properties, have been widely used as a proxy for fractureability. However, brittleness alone does not capture the effect of elevated FIP on fracture geometry.

Role of Minimum Horizontal Stress (Shmin)

(Al-Qasim et al., 2001) and other HPHT case studies highlight that completion component integrity under frac loads is a key limiting factor in deep sour gas wells. Tubing connectors, seals, and packers are especially vulnerable to high differential pressures and thermal expansion effects. Failure to model these loads accurately during design can lead to catastrophic failures during stimulation. Tubing stress analysis tools are used to model load cases, including the "piston effect" where fracture pressure acts across the completion string, generating large axial loads.

Integrated Workflows in Literature

Several authors (Barree et al., 2011; Molinas et al., 2019; Vagenes 2012) recommend a front-end workflow that includes:

1. Detailed mineralogy and elastic property profiling.
2. Shmin and stress anisotropy mapping from logs, image logs, and mini-frac tests.
3. FIP estimation incorporating overpressure and diagenetic effects.
4. Completion load modeling to verify mechanical limits before committing to stimulation designs.

However, industry adoption remains inconsistent, with many projects defaulting to simplified designs due to cost, data gaps, or tight development schedules. The case studies in this paper demonstrate the economic value of implementing a more rigorous, integrated approach.

Case Study 1: Reliance Industries, India

Background

Reliance Industries developed a deep tight gas field offshore India with multiple stacked reservoir intervals. These intervals displayed varying mineralogical compositions and mechanical properties, yet early development plans assumed uniform frackability across the depth range. The deepest wells were expected to be the most productive due to higher total organic content (TOC) and elevated hydrocarbon-filled porosity (HCFP).

Observed Problem

Contrary to predictions, the deepest wells consistently underperformed despite having the highest TOC and HCFP. Initial post-frac production analysis showed lower than expected initial gas rates and more rapid decline compared to shallower offsets. The underperformance prompted a detailed diagnostic investigation to identify controlling factors. Figure 1 shows that, although the deepest wells have the highest HCFP and TOC, their cumulative gas production is markedly lower than shallower wells, revealing that petrophysical quality did not translate to better performance.

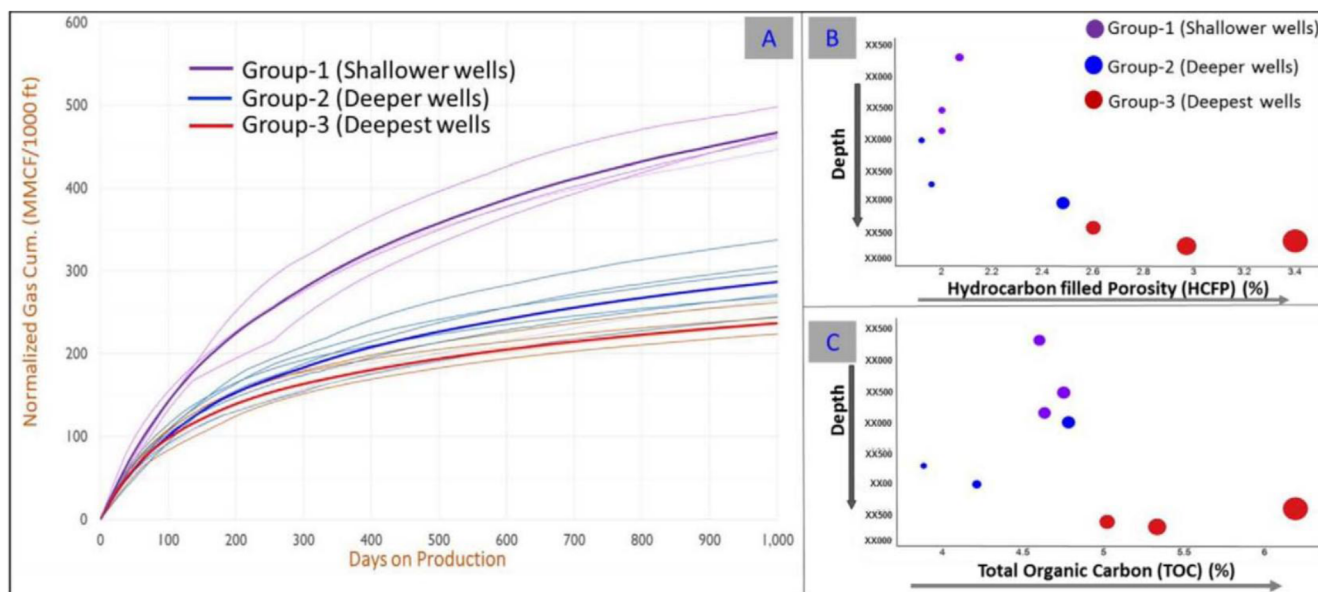


Figure 1—A. cumulative gas production of the three groups of wells based on depth indicating the poor performance of the deepest wells. B. Depth vs. HCFP plot showing highest HCFP in the deepest wells. C. Depth vs. TOC plot showing highest TOC in the deepest wells. (Dutta et al., 2019)

Diagnostic Approach

A multidisciplinary team analyzed core data, petrophysical logs, and frac pressure records.

- **Mineralogy and Elastic Properties:** X-ray diffraction (XRD) data revealed a marked increase in illite and chlorite content with depth, accompanied by a decrease in quartz content. Figure 2 shows Mineral composition trends with depth.
- **Brittleness Index:** Computed brittleness indices showed a downward trend with depth, correlating with increased clay content. Figure 3 shows brittleness vs depth.
- **Sonic-Density Cross-Plots:** These indicated elevated fracture initiation pressures (FIP) in deeper intervals. The cause was attributed to bound water expansion in illitic clays and diagenetic cementation.

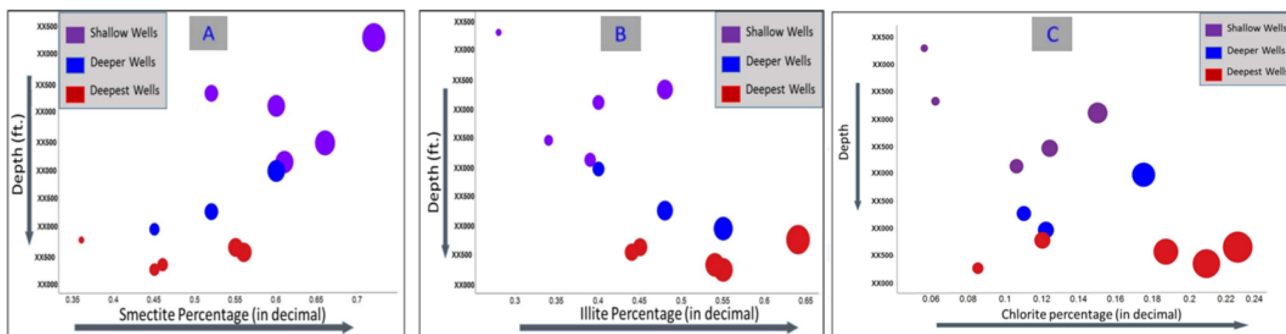


Figure 2—A. Decreasing trend of Smectite% indicating higher diagenesis deeper. B. Increasing trend of illite% indicating higher diagenesis deeper. C. Increasing trend of chlorite% indicating higher diagenesis deeper. (Dutta et al., 2019)

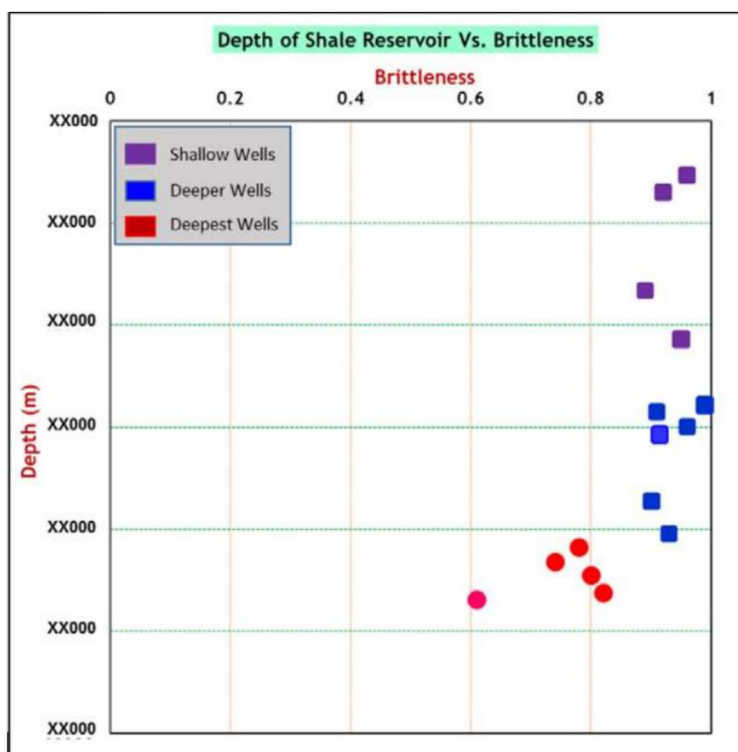


Figure 3—Reservoir Depth vs Brittleness. (Dutta et al., 2019)

Findings

- The higher FIP in deep intervals required higher pump pressures and rates to initiate fractures, increasing operational risk and cost.
- Reduced brittleness limited fracture complexity and proppant placement efficiency.
- Analysis and history matching the hydraulic fracturing treatments performed in the deep wells were performed. Confirming sub-optimal fracture geometry creation. Figure 3 shows Frac Geometry of shallower wells compared to the geometry in deepest wells

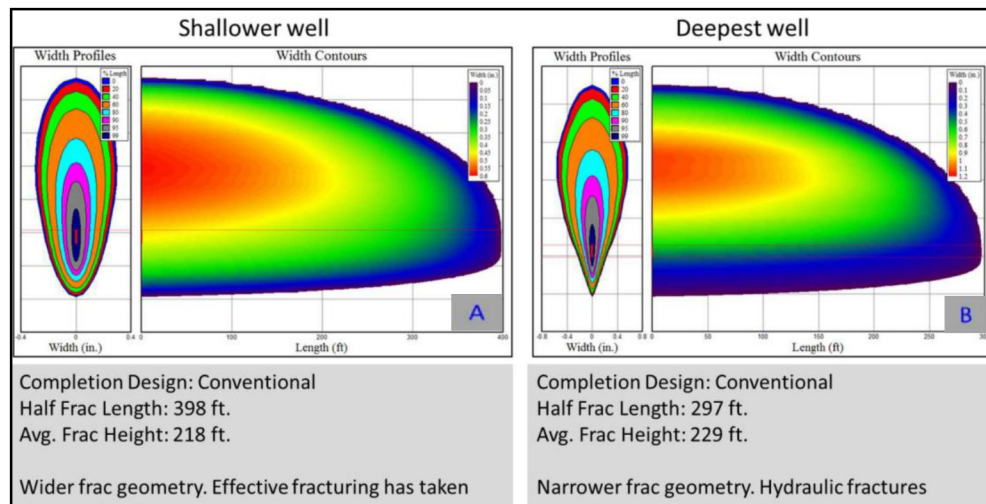


Figure 3a—A. Fracture Geometry in shallower wells. B. Fracture geometry in deepest wells (Dutta et al., 2019)

Solution Implementation

A new frac design is to be tailored for wells that were completed in the deep portion of the reservoir to account for higher FIP and lower brittleness:

- Increased pad volumes to overcome high FIP while controlling net pressure.
- Modified proppant selection to include higher strength materials in deeper intervals.
- Updated stress profiles to inform operational pump limits.

Results

Subsequent wells stimulated with the revised design showed:

- Improved fracture geometry based on microseismic mapping.
- Higher post-frac initial gas rates compared to earlier deep completions.
- Slower production decline rates over the first six months of production.

Case Study 2: Vaca Muerta Shale, Argentina

Background

The Vaca Muerta Shale is one of the most prolific unconventional plays globally, with significant variability in lithology, stress regime, and hydrocarbon-filled porosity (HCFP) across its aerial extent. The operator in this case targeted a development area where earlier appraisal wells had indicated strong production potential in high-HCFP zones.

Observed Anomaly

This is a flipped case to the first case study; as during initial development, the highest-producing well in the campaign was located in a zone with the lowest measured HCFP. This was unexpected, as conventional reservoir screening would have deprioritized this location. In contrast, some high-HCFP wells underperformed. Figure 4 shows production performance of three wells along with reservoir quality of each well

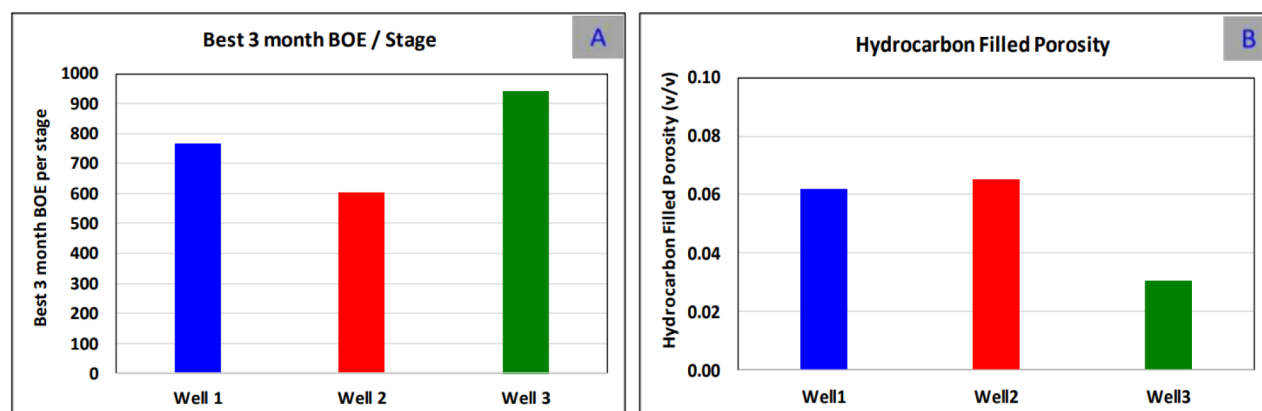


Figure 4—A. Best three months of averaged barrel of oil equivalent (BOE) for three wells. B. Hydrocarbon filled porosity of the three wells (Altman et al., 2018)

Diagnostic Approach

To understand the performance disparity, the operator conducted an integrated review:

- **Petrophysical Evaluation:** Gamma ray, resistivity, and density-neutron logs confirmed the lower porosity in the best well.
- **Geomechanical Analysis:** Image logs and mechanical property derivations revealed that the best well's location corresponded to a zone of unusually low minimum horizontal stress (S_{hmin}). Figure 5 shows minimum horizontal stress values for the three wells
- **Fracture Modeling:** Simulations indicated that in low S_{hmin} zones, fractures extended farther and maintained greater conductivity, offsetting the low HCFP.

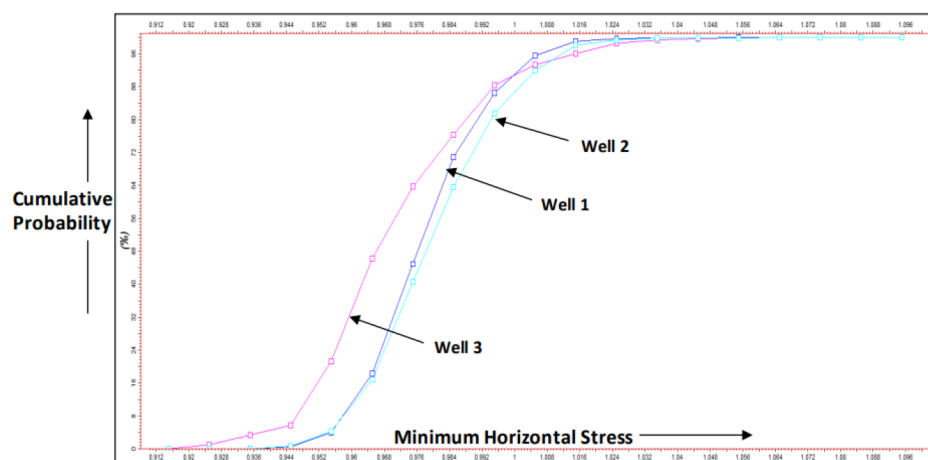


Figure 5—The distribution of minimum horizontal stress values for the three wells. (Altman et al., 2018)

Findings

- Low S_{hmin} was the dominant driver for the superior performance, enabling greater stimulated reservoir volume (SRV) per stage, while high-HCFP wells in higher S_{hmin} zones had lower fracture extension, resulting in smaller SRV and lower production. Table 2 presents the main fracturing geometry parameters derived from history-matching, indicating that Well 3 has a longer fracture half-length relative to the others
- Stress anisotropy mapping was absent from the original well placement strategy.

Table 2—Summarized Key Insights from the Three Case Studies. (Altman et al., 2018)

Parameter	Well 1	Well 2	Well 3
Fracture Half-Length (Xf)	125 m	125 m	178 m
Thickness	47 m	47 m	47 m
Permeability (Fracture cells in SRV)	20 mD	20 mD	20 mD
Fracture Cell Porosity	0.005	0.005	0.002
Sw in Fracture Cells	0.3	0.45	0.6

Solution Implementation

The operator modified the well placement and completion design approach:

- Incorporated Shmin mapping into landing zone selection.
- Adjusted stage spacing and perforation clusters to take advantage of low-stress intervals.
- Optimized hydraulic fracturing design volume. Balanced proppant loading to maintain conductivity in extended fractures.

Results

- Subsequent wells in low-Shmin zones showed 25-40% higher initial production compared to high-HCFP/high-Shmin wells.
- Economic analysis revealed a lower cost per unit of gas produced when low Shmin was targeted, despite lower porosity due to optimized hydraulic treatment design.
- Development in low-Shmin zones enabled accelerated payout.

Case Study 3: Deep Gas, Saudi Arabia

Background

Some formations in Saudi Arabia are deep, high-pressure, high-temperature (HPHT) gas reservoirs containing significant hydrogen sulfide (H₂S) concentrations. Stimulation programs in these formations require hydraulic fracture treatments with surface pressures exceeding 16,000 psi. These extreme conditions place severe demands on completion hardware, particularly tubing, connectors, seals, and packers.

Initial Challenge

Early attempts at fracturing in these formations faced multiple operational constraints:

- High Fracture Initiation Pressure (FIP) due to depth and overburden.
- Material Limitations: Standard completion tubulars and connectors could not withstand the combined stresses from frac pressure, temperature changes, and corrosive H₂S.
- Piston Effect: Large pressure differentials between the inside and outside of the tubing during stimulation generated significant axial loads.

Diagnostic Approach

A comprehensive tubing movement analysis was performed to model the expected loads during stimulation and production. The modeling included:

- Multiple Load Cases: Frac pumping, shut-in, production startup, and kill-weight fluid displacement.

- Thermal Effects: Expansion and contraction of tubulars under HPHT cycles.
- Axial Loads: Quantified from the piston effect and pressure/temperature transients.

Results from the analysis identified that connector failures were the most probable weak point, followed by seal extrusion risks under high differential pressures. Figure 6 shows an example of failure identification from the tubing movement analysis software.

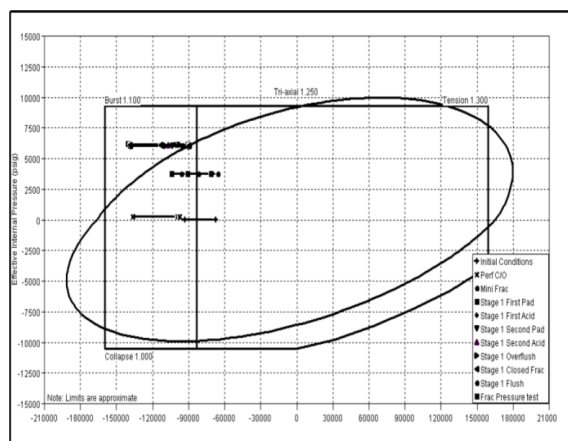


Figure 6—Axial Loads in completion before upgrading indicating working outside of the safe working envelope. (Solares, 2001)

Findings

- The original completion design provided insufficient safety margin for the maximum expected frac load.
- Thermal expansion during production start-up added compressive stresses that compounded axial loads from the piston effect.
- Seal materials were not rated for long-term exposure to high H₂S at elevated temperature

Figure 7 shows tubing movement analysis results with upgraded completion.

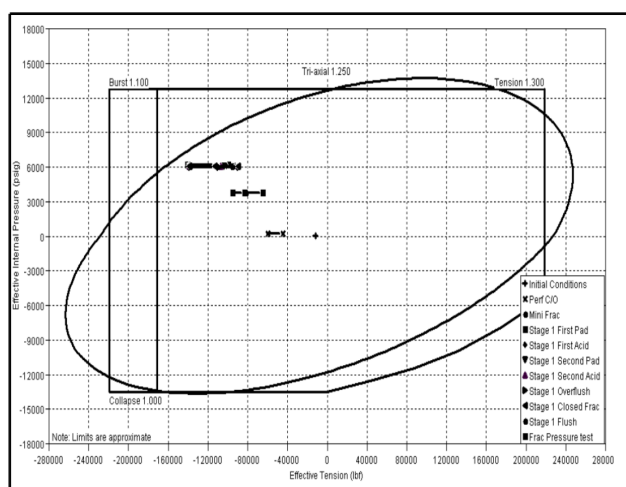


Figure 7—Axial Loads in completion after upgrading indicating working within safe working envelope. (Solares, 2001)

Solution Implementation

The completion design was re-engineered with:

- Upgraded Tubulars: Transition to C-95 and P-110 grades with sour service ratings.
- Connector Redesign: Higher torque capacity connections to handle increased axial and torsional loads.
- Seal Material Upgrade: H₂S-resistant elastomers and back-up rings to prevent extrusion.
- Operational Adjustments: Staged pressurization and controlled temperature ramp-up to minimize thermal shock.

Results

Following implementation of the optimized completion design:

- Fifteen wells were successfully hydraulically fractured without any completion failures.
- Maximum surface treating pressures exceeded 16,000 psi without incident.
- Post-frac production rates met or exceeded pre-job forecasts, validating the mechanical upgrades and operational adjustments.
- Completion upgrades proved economical, as the incremental production compensated for the higher-grade completion costs

Proposed Workflow for Early Detection of Localized Heterogeneity in Developed Reservoir and Optimizing Completion and Stimulation Design

In developed fields, well performance generally follows predictable patterns based on extensive historical data. However, some wells despite being located near consistent offset producers exhibit unexpected underperformance or, in rarer cases, overperformance. Such deviations often trace back to localized heterogeneity in geomechanical, petrophysical, or completion-related constraints that went unrecognized during initial field development planning.

To address this, the proposed workflow is designed to identify and characterize such isolated anomalies before committing to final completion and stimulation designs.

Step 1 – Early Data Integration

Aggregate logs, core, image logs, and offset well production into a combined dataset during well planning. Explicitly look for outlier signatures that differ from the established field model.

Step 2 – Petrophysical and Mineralogical Profiling

- Calculate total organic content (TOC) and hydrocarbon-filled porosity (HCFP).
- Identify mineralogical trends with depth, noting changes in brittle minerals (quartz, feldspar, carbonates) versus ductile clays.
- Map zones of high brittleness to prioritize for stimulation.

Step 3 – Geomechanical Modeling

- Determine in-situ stress magnitudes and directions (S_{hmin} , S_{Hmax} , S_v) from log interpretation, image logs, and mini-frac tests.
- Map stress anisotropy and correlate with reservoir quality indicators.

- Estimate fracture initiation pressure (FIP) using combined mechanical and pore pressure models.

Step 4 – Adaptive Stimulation Design & Optimization

Tailor stimulation parameters to the localized condition:

- High FIP / ductile intervals increase pad volume, rate, and proppant intensity.
- Low-stress intervals reduce injected volume and horsepower while still achieving optimum SRV, as demonstrated in Case 2.

Step 5 – Completion Mechanical Integrity Verification

Once frac design parameters are known, conduct load-case modeling to confirm the completion hardware (tubulars, connectors, seals) can safely withstand predicted pressures, thermal changes, and corrosive environments. Modify hardware specifications if necessary before execution.

Step 6 – Execution and Real-Time Monitoring

Monitor fracture geometry and bottomhole pressures to validate assumptions and make on-the-fly adjustments to pumping schedules or stage designs.

Step 7 – Post-Job Evaluation and Feedback Loop

Incorporate results into the field development model to refine the prediction of localized anomalies and optimize designs for future wells in similar settings.

Discussion

These three case studies illustrate that even in well-developed fields, where reservoir characterization is mature and offset well performance is predictable, localized anomalies can still occur and significantly influence outcomes. Such anomalies may be detected late after the well is drilled or even after early production if planning relies solely on field-average parameters.

In each case:

Case 1 (India) – Despite high TOC and HCFP, certain intervals exhibited elevated fracture initiation pressures and lower brittleness, limiting fracture complexity and conductivity. These zones required more aggressive stimulation parameters than the standard field design provided.

Case 2 (Argentina) – Low-stress conditions, while associated with lower HCFP, enabled rapid and extensive SRV development with reduced pumping effort. This presented a unique opportunity to downsize the frac design lower volume, lower rate, and reduced horsepower without sacrificing SRV or production, conserving both capital and resources.

Case 3 (Saudi Arabia) – Completion hardware became the controlling constraint. HPHT and sour service conditions demanded design adjustments to avoid overstressing tubulars and connections during stimulation.

Importance of Design-Before-Completion

A key lesson is that tailored frac design must precede completion hardware selection. In both high- demand scenarios (e.g., Case 1) and optimized low-demand scenarios (e.g., Case 2), knowing the stimulation loads first ensures that the completion system is neither over- nor under-specified. This sequencing avoids costly redesigns, prevents operational limits from compromising stimulation objectives, and ensures mechanical integrity under actual job conditions

Broader Implications

1. Localized heterogeneity detection is essential – Even in developed fields, targeted diagnostics can uncover well-specific geomechanical or petrophysical differences.
2. Offset well data is not always predictive; Anomalies may arise from stratigraphic variations, local stress changes, or mechanical constraints invisible in field-wide models.
3. Early integration of design and hardware verification improves outcomes – By determining frac loads first, operators can optimize both stimulation effectiveness and completion longevity.
4. Optimization opportunities are as valuable as remediation – Case 2 demonstrates that recognizing favorable stress conditions can lead to smaller, more efficient stimulations that maintain performance while lowering environmental footprint and operational costs.

Completion Mechanical Limits as a Hidden Constraint

The Saudi case underscores a challenge often neglected in early planning: the mechanical limits of tubulars, connectors, and seals under combined frac and thermal loads. Even when geomechanical and petrophysical conditions are favorable, failure to verify completion integrity can render an optimal stimulation design impossible to execute.

Conclusions & Recommendations

Conclusions

Even in well-developed gas fields with a long history of consistent production, certain wells may behave very differently from their nearby offsets due to localized variations in geomechanical, petrophysical, or operational conditions. These anomalies whether they manifest as underperformance in apparently high-quality rock or overperformance in less favorable petrophysical settings are often detected only after drilling or early production. This late recognition points to the importance of proactively searching for localized heterogeneity, even in mature developments.

The three field cases demonstrate that stimulation strategies must be adaptable to these localized realities. In wells where rock and stress conditions are unfavorable for fracture propagation, such as in Case 1, achieving effective SRV requires larger stimulation volumes and higher pumping rates than the field's standard frac design. Conversely, in favorable low-stress conditions such as in Case 2, the same SRV can be achieved with significantly smaller jobs lower rates, lower volumes, and reduced horsepower without compromising productivity. In high-pressure, high-temperature environments such as in Case 3, completion mechanical limits may dictate the operational envelope, regardless of reservoir quality.

Across these scenarios, one consistent lesson emerges: the frac design must be established before specifying the completion system, ensuring the hardware can accommodate the predicted loads while avoiding unnecessary overdesign. Applying this integrated approach can increase capital efficiency, reduce operational risk, and improve sustainability in both challenging and conventional environments.

Recommendations

To improve performance predictability and optimize stimulation effectiveness in mature fields, operators should integrate localized heterogeneity detection into the earliest planning phases. This requires combining offset performance analysis with high-resolution logs, core evaluation, and geomechanical modeling to identify intervals with atypical fracture initiation pressures, stress states, or brittleness. Stress mapping should be considered as important as petrophysical evaluation, ensuring both are incorporated into frackability assessments.

Stimulation design should always be defined before finalizing completion hardware, enabling load-case analysis to confirm mechanical integrity under predicted job conditions. For wells with unfavorable

frackability, operators should be prepared to exceed standard field volumes and rates to achieve sufficient SRV. For wells in favorable low-stress settings, they should instead optimize by reducing job size, rate, and horsepower, conserving water, proppant, fuel, and costs while maintaining production levels. Real-time fracture monitoring should be employed during execution to confirm model assumptions and allow parameter adjustments on the fly. Finally, post-job results, especially from anomalous wells should be fed back into the development plan, ensuring that future wells benefit from these lessons and reducing the likelihood of late-stage surprises.

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Nomenclature

TOC	= Total Organic Content
HCFP	= Hydrocarbon Filled Porosity
FIP	= Fracture Initiation Pressure
SRV	= Stimulated Reservoir Volume
HPHT	= High Pressure High Temperature
EUR	= Estimated Ultimate Recovery
UCS	= Unconfined Compressive Strength
XRD	= X-Ray Diffraction
PAD Volume	= Volume of clean fracturing fluid pumped before proppant-laden stages to initiate and extend fractures.
Net Pay	= Portion of a Reservoir That Contains Economically Producing Hydrocarbons
CFD	= Computational Fluid Dynamics

Subscripts

S_{hmin}	= Minimum Horizontal Stress
S_{Hmax}	= Maximum Horizontal Stress
S_v	= Overburden Stress

References

1. Barree, R.D., Barree, V.L., Craig, D.P. 2011. Holistic Fracture Diagnostics: Consistent Interpretation of Prefrac Injection Tests Using Multiple Analysis Methods. Paper presented at the SPE Annual Technical Conference and Exhibition, Denver, Colorado, 30 October-2 November. SPE-146776-MS. <https://doi.org/10.2118/146776-MS>
2. Molinas, M., Perez, R., Varela, J., et al 2019. Challenges and Learnings from Hydraulic Fracturing in Unconventional Reservoirs: Vaca Muerta Case Study. Paper presented at the SPE Latin American and Caribbean Petroleum Engineering Conference, Buenos Aires, Argentina, 17-19 July. SPE-194657-MS. <https://doi.org/10.2118/194657-MS>
3. Riazi, N., Weijers, L., Wilson, B., et al 2017. Unconventional Reservoir Completions: Lessons Learned from Multidisciplinary Data Integration in the Vaca Muerta Shale. Paper presented at the Unconventional Resources Technology Conference, Austin, Texas, 24-26 July. URTeC-2902306. <https://doi.org/10.15530/urtec-2017-2902306>

4. Al-Qasim, A., Al-Nuaim, S., Al-Ghamdi, A., et al 2001. Design and Execution of Hydraulic Fracturing Treatments in HPHT Sour Gas Wells in Saudi Arabia. Paper presented at *the SPE Middle East Oil Show*, Bahrain, 17-20 March. SPE-68206-MS. <https://doi.org/10.2118/68206-MS>
5. Vagenes, E. 2012. *Understanding Fracability in Unconventional Reservoirs: The Importance of Integrated Analysis*. MSc Thesis, Norwegian University of Science and Technology (NTNU), Trondheim, Norway.
6. Reliance Industries Limited. 2018. Field Development Summary and Post-Frac Evaluation Reports – Deep Tight Gas Wells Offshore India. *Internal Technical Report*.